

Chapter 2: Antenna Fundamentals

Fields, Radiation Patterns, Directivity, Gain, Impedance, Efficiency, and Polarization

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Stutzman & Thiele, *Antenna Theory and Design*, 3rd Edition



Lecture Roadmap

- ① Maxwell equations in phasor form
- ② Vector potential and radiation fields
- ③ Ideal dipole: fields and pattern
- ④ Near field, radiating near field, far field
- ⑤ Radiation pattern and HPBW
- ⑥ Directivity, gain, impedance, efficiency
- ⑦ Polarization and mismatch

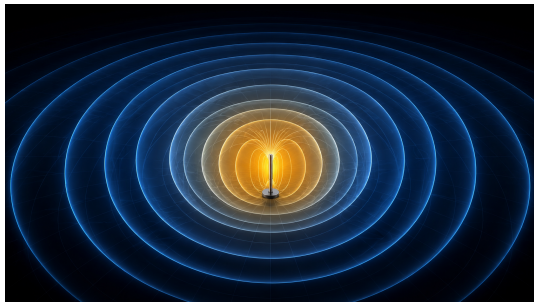
Central question

If the antenna current is known, how do we compute the radiated field and antenna performance?

$$J \rightarrow A \rightarrow E, H \rightarrow F, D, G, Z_A, e_r$$



Why Chapter 2 Matters



- Chapter 1: qualitative view of antennas and radiation
- Chapter 2: quantitative field and parameter framework
- The antenna becomes both:
 - an electromagnetic radiator
 - an RF circuit load

Engineering message

Antenna performance is not one number. It is a coupled set of field, circuit, and system properties.



Time-Harmonic Field Assumption

Most antenna problems are analyzed at a single carrier frequency.

$$\mathcal{E}(\mathbf{r}, t) = \Re\{E(\mathbf{r})e^{j\omega t}\}$$

$$\frac{\partial}{\partial t} \longrightarrow j\omega$$

- Phasors remove explicit time dependence
- Magnitude and phase are retained
- Same idea as sinusoidal steady-state circuit analysis

Key constants

$$\beta = \omega\sqrt{\mu\epsilon}$$

$$\beta = \frac{2\pi}{\lambda}, \quad \lambda = \frac{c}{f}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$



Maxwell Equations in Phasor Form

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} + \mathbf{J}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}$$

$$\nabla \cdot \mathbf{H} = 0$$

$$\nabla \cdot \mathbf{J} = -j\omega\rho$$

Interpretation

- Time-varying electric and magnetic fields create each other
- Source current density $\mathbf{J}$ drives radiation
- Charge conservation is enforced by continuity

Antenna viewpoint

The impressed current on the antenna is the source of the radiated field.

Power Flow: The Poynting Vector

$$S = \frac{1}{2} E \times H^*$$

$$P = \Re \left\{ \iint_s S \cdot ds \right\}$$

- S : complex power density
- $\Re\{S\}$: time-average real power density
- Surface integral gives total power flow

Radiation test

In the far field, power flows radially outward and does not return to the antenna.

$$E, H \propto \frac{1}{r}$$

$$S \propto \frac{1}{r^2}$$



Vector Potential Formulation

Instead of solving directly for E and H , define

$$H = \frac{1}{\mu} \nabla \times A$$

Under the Lorenz condition,

$$\nabla^2 A + \beta^2 A = -\mu J$$

For a general source,

$$A = \iiint_{V'} \mu J(\mathbf{r}') \frac{e^{-j\beta R}}{4\pi R} dv'$$

Physical meaning

Each small current element contributes a delayed spherical wave.

$$\frac{1}{R} \rightarrow \text{spreading}$$

$$e^{-j\beta R} \rightarrow \text{phase delay}$$



Far-Field Approximation

For a source point $\mathbf{r}'$ and observation point far away:

$$R \approx r - \hat{\mathbf{r}} \cdot \mathbf{r}'$$

Then

$$A \approx \mu \frac{e^{-j\beta r}}{4\pi r} \iiint_{V'} J(\mathbf{r}') e^{j\beta \hat{\mathbf{r}} \cdot \mathbf{r}'} dV'$$

Far-field properties

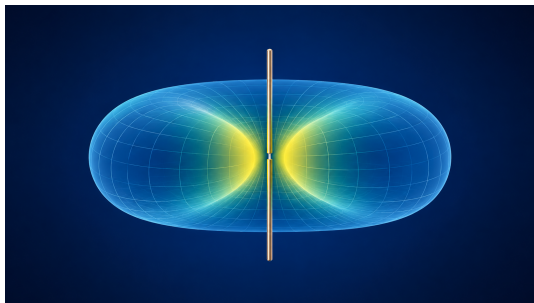
$$E = -j\omega(A_\theta \hat{\theta} + A_\phi \hat{\phi})$$

$$H = \frac{1}{\eta} \hat{\mathbf{r}} \times E$$

$$E \perp H \perp \hat{\mathbf{r}}$$



The Ideal Dipole



- A very short current element
- Length condition:

$$\Delta z \ll \lambda$$

- Current assumed uniform:

$$I(z) = I$$

- Also called Hertzian dipole or current element

Why it matters

It is the elementary radiator used to build intuition for all antennas.

Far Fields of the Ideal Dipole

For a z-directed ideal dipole:

$$E_{\theta} = j\omega\mu \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin \theta$$

$$H_{\phi} = j\beta \frac{I\Delta z}{4\pi r} e^{-j\beta r} \sin \theta$$

$$\frac{E_{\theta}}{H_{\phi}} = \frac{\omega\mu}{\beta} = \eta$$

$$F(\theta) = \sin \theta$$

Interpretation

The ideal dipole radiates maximum field perpendicular to its axis and zero field along its axis.



Numerical Example 1: Ideal Dipole Fields

Given:

$$\begin{aligned}f &= 300 \text{ MHz}, \quad \lambda = 1 \text{ m} \\ \Delta z &= \frac{\lambda}{50} = 0.02 \text{ m}, \quad I = 1 \text{ A} \\ r &= 100 \text{ m}, \quad \theta = 90^\circ\end{aligned}$$

Results:

$$\begin{aligned}|E_\theta| &= 0.0377 \text{ V/m} \\ |H_\phi| &= 1.0 \times 10^{-4} \text{ A/m} \\ S_{av} &= 1.89 \times 10^{-6} \text{ W/m}^2 \\ R_r &= 0.316 \Omega\end{aligned}$$

Lesson

A very small antenna can have extremely small radiation resistance.



Field Regions Around an Antenna

Reactive near field

Stored energy dominates; for small antennas,

$$r \lesssim \frac{\lambda}{2\pi}.$$

Radiating near field

Radiation exists, but the pattern still changes with distance.

Far field

Pattern depends only on angle; fields behave as

$$E, H \propto \frac{1}{r}.$$

Large-antenna criterion

$$r_{ff} \approx \frac{2D^2}{\lambda}$$

Measurement warning

A pattern measured too close is not the true far-field pattern.



Radiation Pattern Definitions

Normalized field pattern:

$$F(\theta, \phi) = \frac{E(\theta, \phi)}{E_{max}}$$

Normalized power pattern:

$$P(\theta, \phi) = |F(\theta, \phi)|^2$$

In decibels:

$$20 \log_{10} |F| = 10 \log_{10} |F|^2$$

- Main lobe
- Side lobes
- Back lobe
- Nulls
- Half-power beamwidth

Half-power point

$$|F| = \frac{1}{\sqrt{2}} = 0.707$$

$$P = -3 \text{ dB}$$



Numerical Example 2: Pattern and HPBW

For the ideal dipole:

$$F(\theta) = \sin \theta, \quad P(\theta) = \sin^2 \theta$$

θ	$F(\theta)$	$P(\theta)$
0°	0	0
30°	0.500	0.250
45°	0.707	0.500
90°	1.000	1.000
135°	0.707	0.500
180°	0	0

$$\text{HPBW} = 135^\circ - 45^\circ = 90^\circ$$



Radiation from Line Currents

For a z-directed line current:

$$A_z = \mu \frac{e^{-j\beta r}}{4\pi r} \int I(z') e^{j\beta z' \cos \theta} dz'$$

$$E_\theta = j\omega A_z \sin \theta$$

Key idea

The radiation pattern is the coherent far-field sum of all current elements on the antenna.

Why phase matters

Small path-length differences become angular phase differences, shaping beams and nulls.



Element Factor and Pattern Factor

Many patterns can be written as

$$F(\theta, \phi) = g(\theta, \phi)f(\theta, \phi)$$

Element factor

Pattern of an infinitesimal current element.

$$g(\theta) = \sin \theta$$

for a z-directed current element.

Pattern factor

Effect of source distribution and phase summation.

$$f(\theta, \phi)$$

$$F_{total} = F_{element} F_{array}$$



Directivity

Radiation intensity:

$$U(\theta, \phi) = r^2 S(\theta, \phi)$$

Average radiation intensity:

$$U_{ave} = \frac{P_{rad}}{4\pi}$$

Maximum directivity:

$$D = \frac{U_{max}}{U_{ave}} = \frac{4\pi U_{max}}{P_{rad}}$$

Beam solid angle:

$$\Omega_A = \iint_{4\pi} |F(\theta, \phi)|^2 d\Omega$$

$$D = \frac{4\pi}{\Omega_A}$$



Numerical Example 3: Directivity and Gain

For an ideal dipole:

$$F(\theta) = \sin \theta$$

$$\Omega_A = \int_0^{2\pi} \int_0^\pi \sin^2 \theta \sin \theta d\theta d\phi = \frac{8\pi}{3}$$

$$D = \frac{4\pi}{8\pi/3} = 1.5$$

$$D_{dBi} = 10 \log_{10}(1.5) = 1.76 \text{ dBi}$$

If $e_r = 0.85$:

$$G = e_r D = 0.85(1.5) = 1.275$$

$G = 1.06 \text{ dBi}$



Gain versus Directivity

$$G = e_r D$$

$$0 \leq e_r \leq 1$$

$$G_{dB} = 10 \log_{10} G$$

$$D_{dB} = 10 \log_{10} D$$

Directivity

How strongly radiation is concentrated in angle.

Gain

How strongly useful power is radiated, including losses.

Important

A narrow beam with poor efficiency may not produce high gain.

Antenna Input Impedance

From the transmitter or receiver viewpoint, the antenna is a load:

$$Z_A = R_A + jX_A$$

$$R_A = R_r + R_o$$

Real part

R_r : radiated power

R_o : ohmic loss

Imaginary part

X_A : stored near-field energy

Capacitive or inductive behavior.

$$P_{in} = \frac{1}{2} R_A |I_A|^2$$



Radiation Resistance

Definition:

$$R_r = \frac{2P_{rad}}{|I_A|^2}$$

For an ideal dipole:

$$R_r = 80\pi^2 \left(\frac{\Delta z}{\lambda} \right)^2 \Omega$$

For a practical short dipole with triangular current:

$$R_r = 20\pi^2 \left(\frac{L}{\lambda} \right)^2 \Omega$$

Engineering lesson

As antenna size becomes much smaller than wavelength, radiation resistance becomes very small.

Numerical Example 4: Short Dipole Impedance

Given:

$$f = 30 \text{ MHz}, \quad L = 0.5 \text{ m}, \quad R_o = 0.20 \Omega, \quad X_A = -600 \Omega$$

$$\lambda = \frac{3 \times 10^8}{30 \times 10^6} = 10 \text{ m}, \quad \frac{L}{\lambda} = 0.05$$

$$R_r = 20\pi^2(0.05)^2 = 0.494 \Omega$$

$$R_A = R_r + R_o = 0.694 \Omega$$

$$e_r = \frac{R_r}{R_r + R_o} = 0.712$$

$$Z_A = 0.694 - j600 \Omega$$



Efficiency and Matching Are Different

Radiation efficiency:

$$e_r = \frac{R_r}{R_r + R_o}$$

Mismatch coefficient:

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0}$$

Accepted power fraction:

$$1 - |\Gamma|^2$$

For the short dipole example with $Z_0 = 50\Omega$:

$$|\Gamma| \approx 0.9998$$

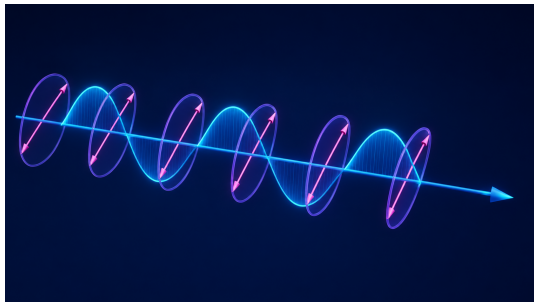
$$1 - |\Gamma|^2 \approx 3.83 \times 10^{-4}$$

Only about 0.038% of incident power is accepted without matching.

Design implication

A small antenna may require a matching network even if its radiation efficiency is acceptable.

Polarization



Polarization describes the time path of the electric field vector tip.

$$\mathbf{E} = E_x \hat{x} + E_y \hat{y}$$

$$E_x = E_1 \cos \omega t$$

$$E_y = E_2 \cos(\omega t + \delta)$$

- Linear
- Circular
- Elliptical



Numerical Example 5: Polarization Mismatch

For two linearly polarized antennas:

$$\text{PLF} = \cos^2 \psi$$

If the receive antenna is tilted by

$$\psi = 45^\circ$$

then

$$\text{PLF} = \cos^2(45^\circ) = \left(\frac{1}{\sqrt{2}}\right)^2 = 0.5$$

Power loss:

$$10 \log_{10}(0.5) = -3.01 \text{ dB}$$

Result

A 45° polarization mismatch loses half the available received power.

Linear, Circular, and Elliptical Cases

Case	E_1	E_2	δ
Linear	nonzero	0 or in phase	0° or 180°
Circular	equal	equal	$\pm 90^\circ$
Elliptical	general	general	general

Example:

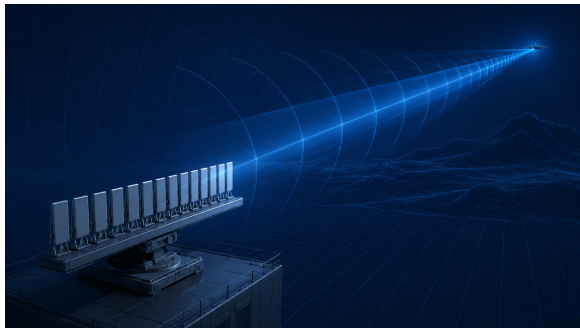
$$E_1 = 2, \quad E_2 = 1, \quad \delta = 90^\circ$$

$$\boxed{\text{elliptical polarization}}, \quad AR = \frac{2}{1} = 2 = 6.02 \text{ dB}$$

System use

Polarization can cause loss, but it can also provide an additional diversity channel.

Connection to Radar Systems



- Gain increases transmitted power density toward the target
- Receive gain improves echo collection
- Polarization affects target scattering response
- Pattern controls angular coverage and clutter illumination

Radar implication

Antenna parameters enter directly into link budget and radar equation performance.



$$F_{total}(\theta, \phi) = F_{element}(\theta, \phi)F_{array}(\theta, \phi)$$

received phase $\rightarrow$ steering vector

steering vector $\rightarrow$ array covariance

array covariance $\rightarrow$ MUSIC/ESPRIT

Common modeling risk

If the element pattern, polarization, or coupling is ignored, the array manifold becomes distorted.

- Biased DOA estimates
- Weaker nulls
- Reduced beamforming gain



Connection to FPGA/DSP Implementation

Analog electromagnetic front end

- Pattern
- Gain
- Polarization
- Matching
- Efficiency

Digital processing chain

- ADC samples
- Filtering
- Covariance estimation
- Beamforming / DOA
- FPGA acceleration

Antenna response $\rightarrow$ received samples $\rightarrow$ DSP/FPGA result

Core message

Digital algorithms cannot recover information that the antenna front end failed to capture.

Teaching Summary: What Students Must Remember

- ① Current distribution is the source of radiation.
- ② Far fields are transverse and behave locally like plane waves.
- ③ The ideal dipole has $F(\theta) = \sin \theta$.
- ④ Directivity is pattern concentration; gain includes efficiency.
- ⑤ Input impedance combines radiation, ohmic loss, and stored energy.
- ⑥ Radiation efficiency and impedance matching are not the same.
- ⑦ Polarization mismatch can cause real link loss.



$$J \rightarrow A \rightarrow E, H \rightarrow F(\theta, \phi)$$

$$F(\theta, \phi) \rightarrow D$$

$$D, e_r \rightarrow G$$

$$R_r, R_o, X_A \rightarrow Z_A, e_r$$

Engineering definition

An antenna is an electromagnetic transducer that converts guided electrical energy into radiated waves, and vice versa.

Chapter 2 foundation

This chapter prepares us for dipoles, monopoles, arrays, aperture antennas, measurements, radar links, and DOA systems.

